

DATA STRUCTURES AND ALGORITHMS

LECTURE NO 18

Trees I

SQUADRON LEADER (RETD) ABDUL REHMAN

RECALL

Arrays

Vectors

Linked
Lists

Stacks

Queues

- Trees
- Graphs

TREES

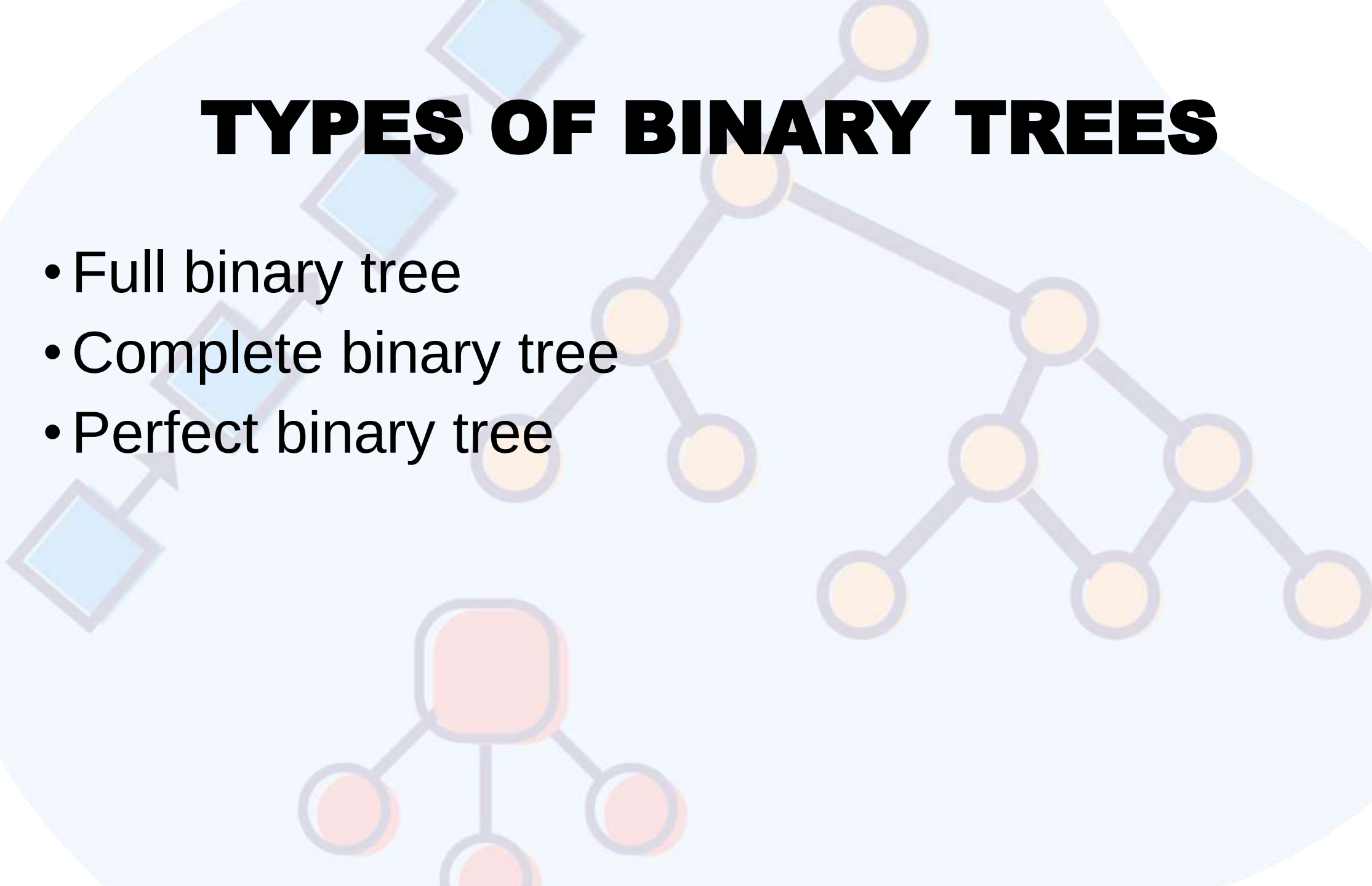
- A tree is a nonlinear data structure that consists of nodes connected by edges.
- It is defined as a collection of nodes, where each node can have zero or more child nodes
- There is exactly one root node with no parent

BINARY TREES

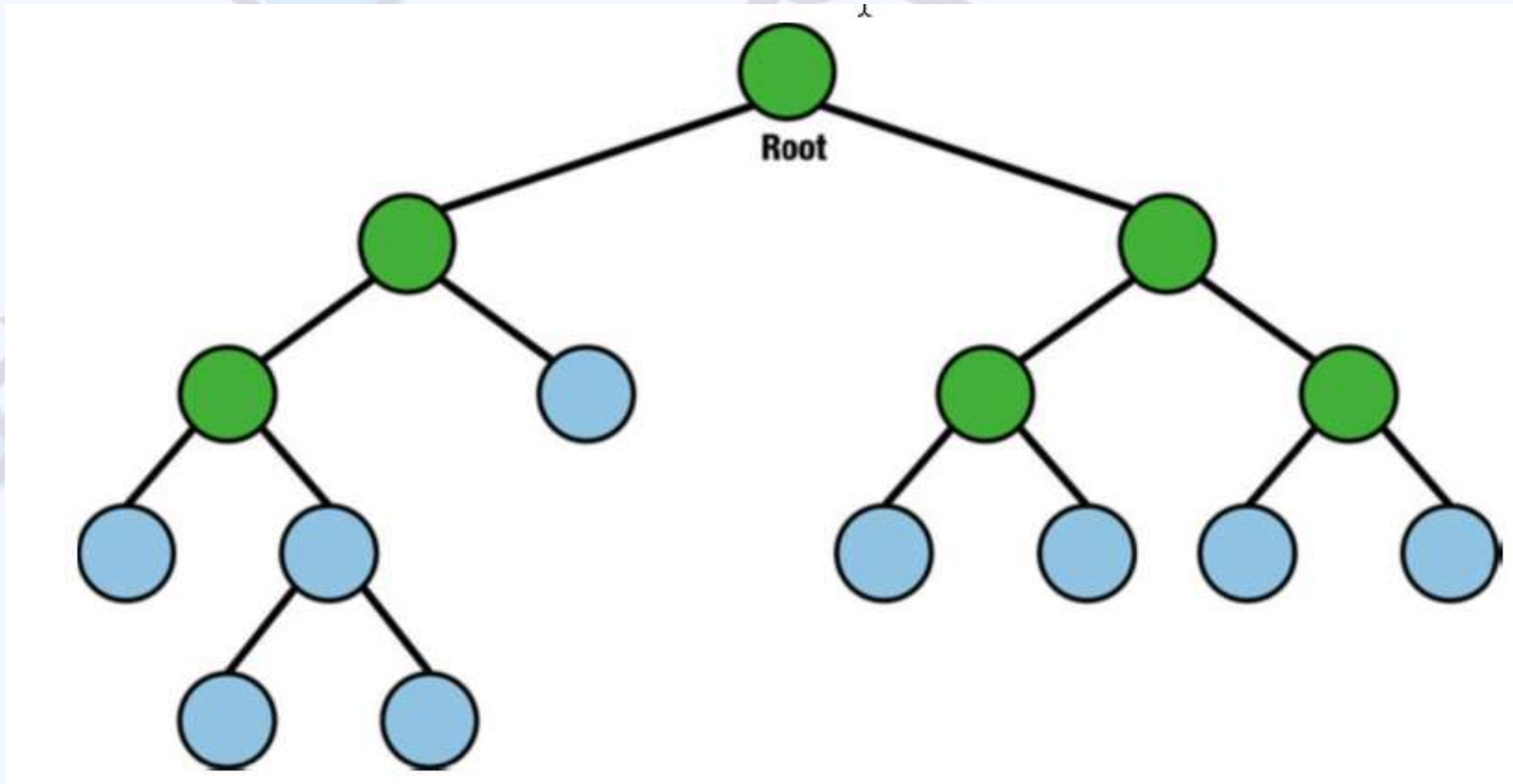
- Binary trees are a type of data structure where each node has at most two children, referred to as
 - Left child
 - Right child
- This structure allows for efficient data storage retrieval and manipulation
- Binary trees are used in various applications, such as
 - Expression parsing
 - Decision-making processes
 - Data sorting.

TYPES OF BINARY TREES

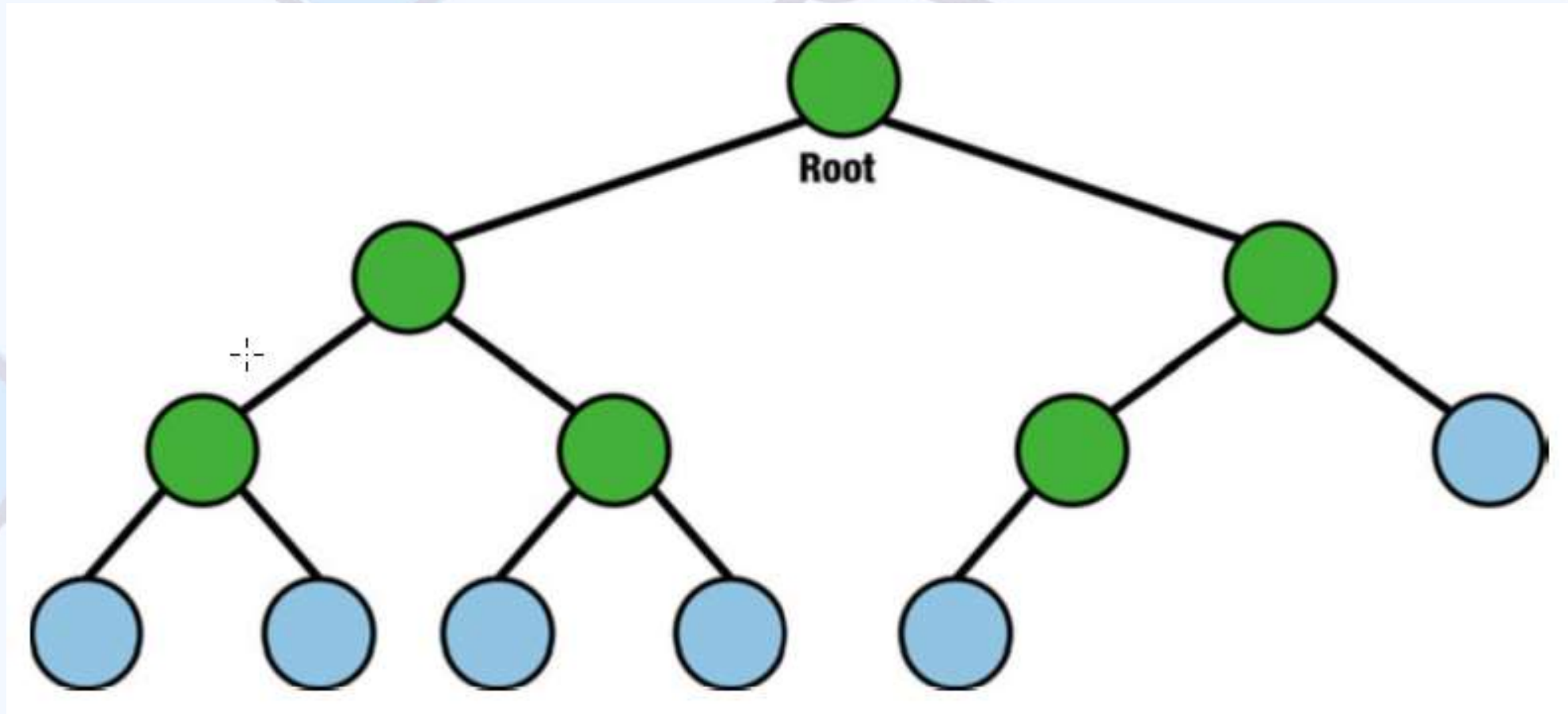
- Full binary tree
- Complete binary tree
- Perfect binary tree



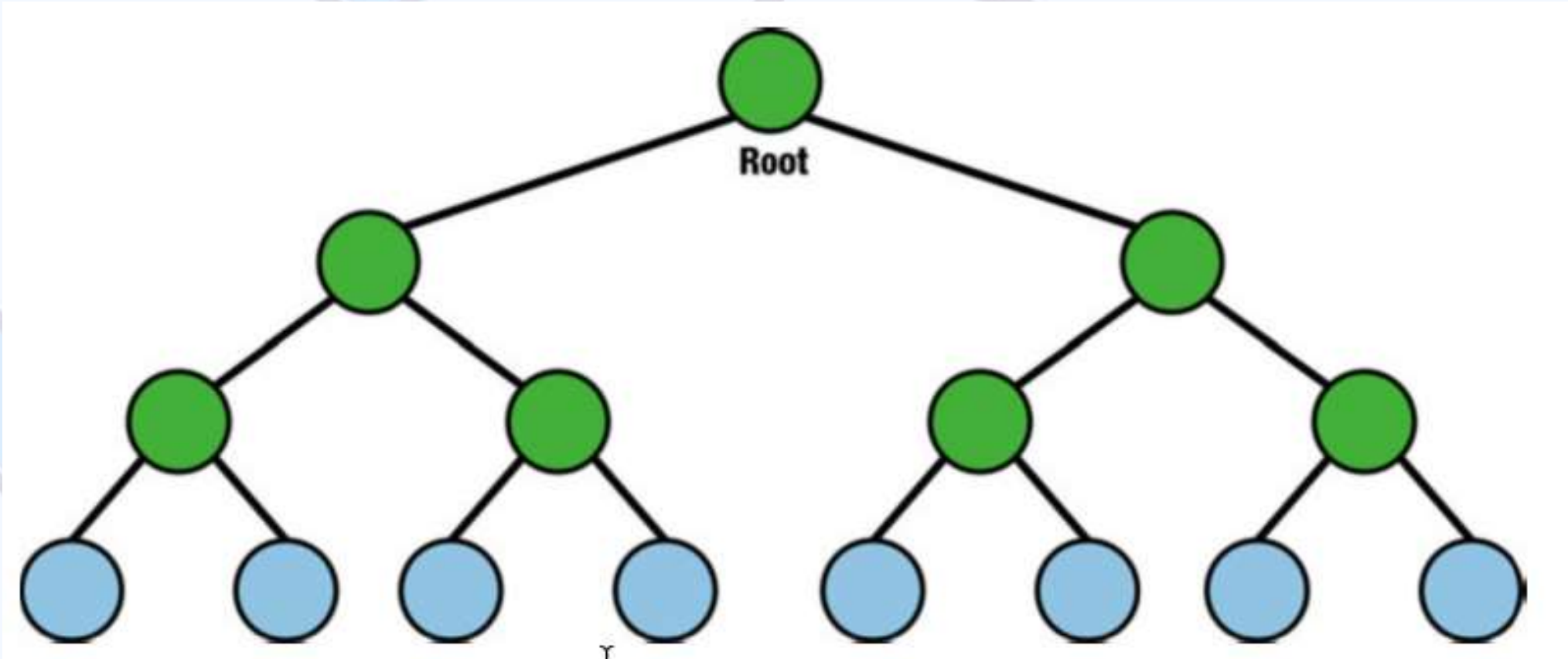
FULL BINARY TREE



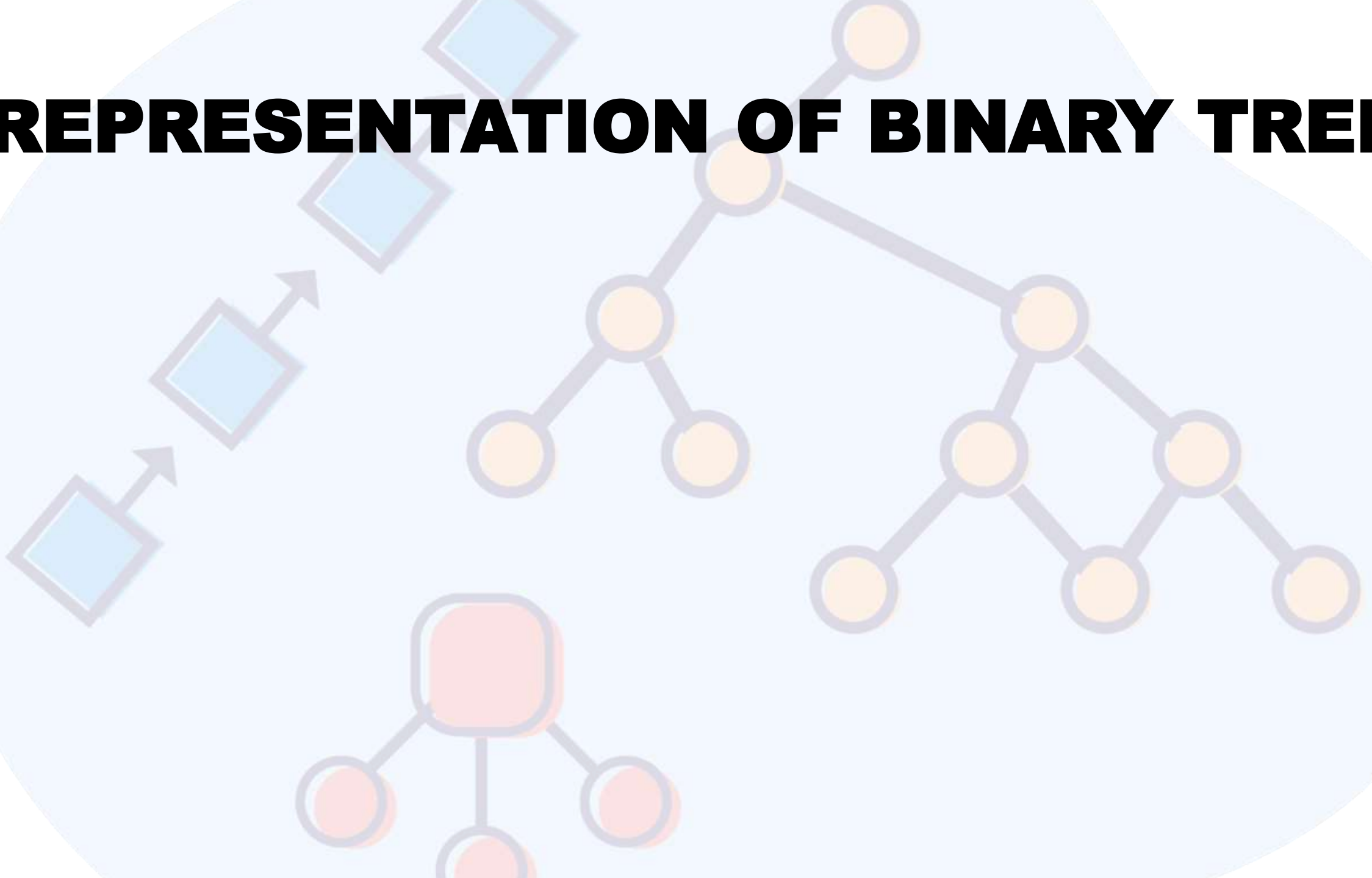
COMPLETE BINARY TREE



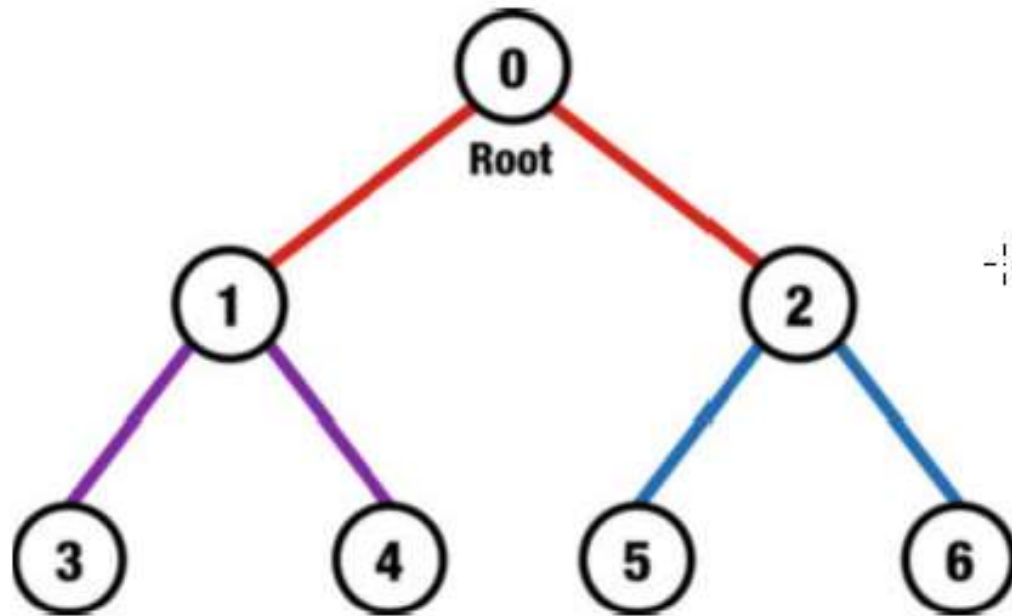
PERFECT BINARY TREE



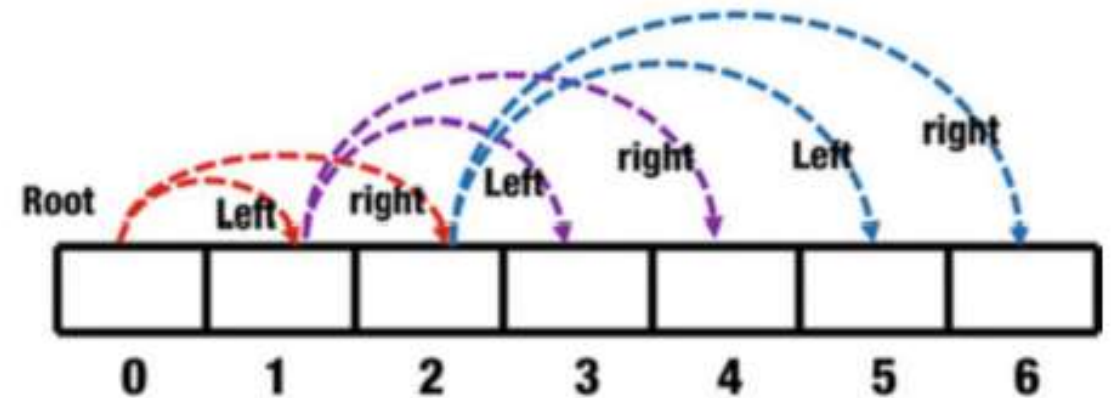
REPRESENTATION OF BINARY TREE



REPRESENTATION OF BINARY TREE

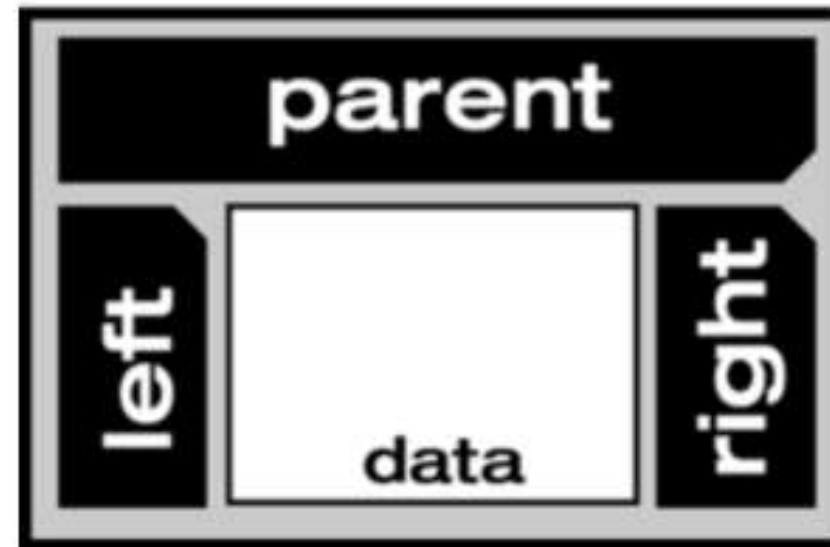


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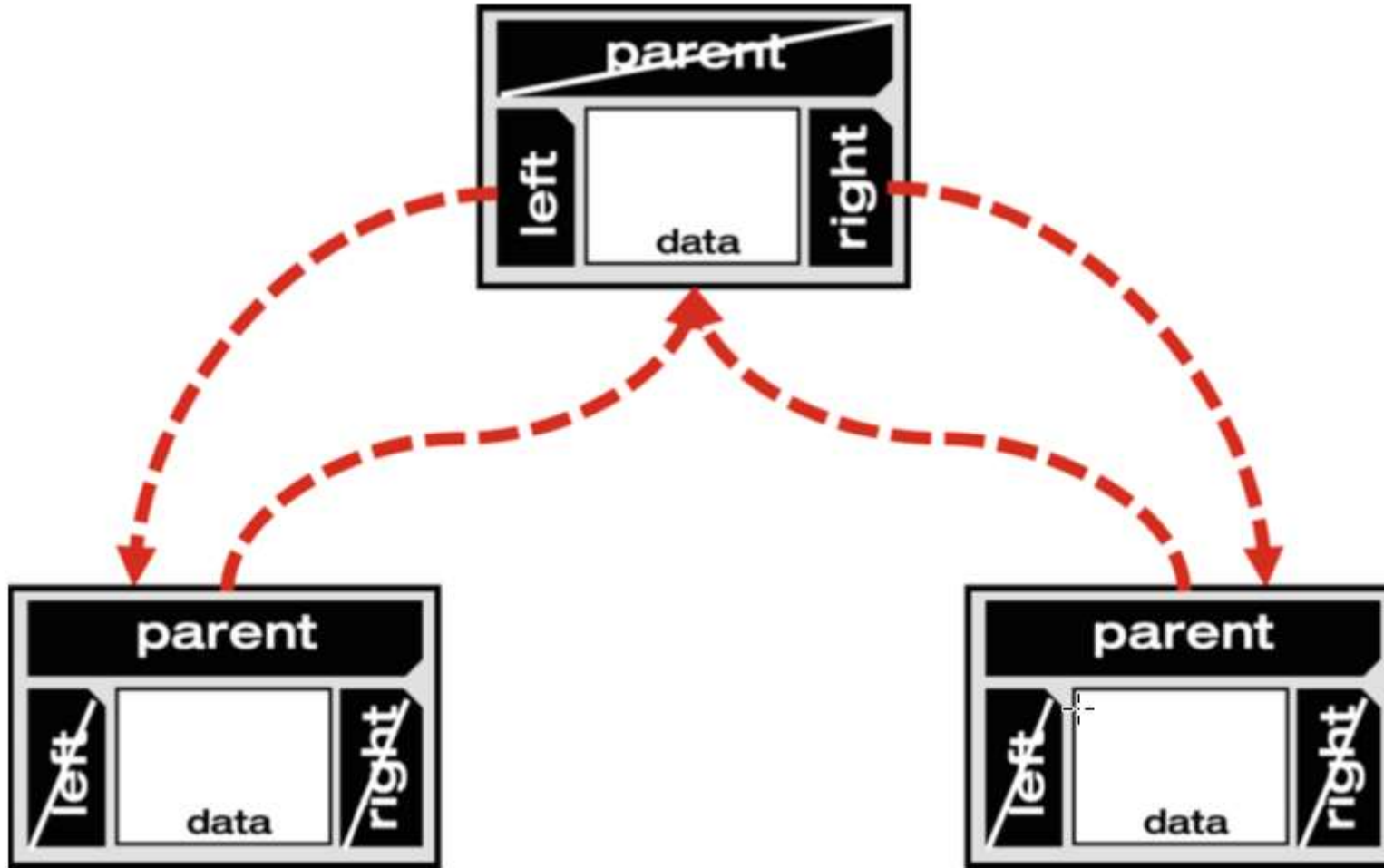


THROUGH LINKED LIST

Node



THROUGH LINKED LIST



- Unlike Heaps
 - **which use array index math $2i+1$**
- General Trees are built using dynamic memory and pointers.
 - Can you recall any such structure?



Data Structures in Depth Using C++

A Comprehensive Guide to Data Structure
Implementation and Optimization in C++

Mahmoud Mahdi